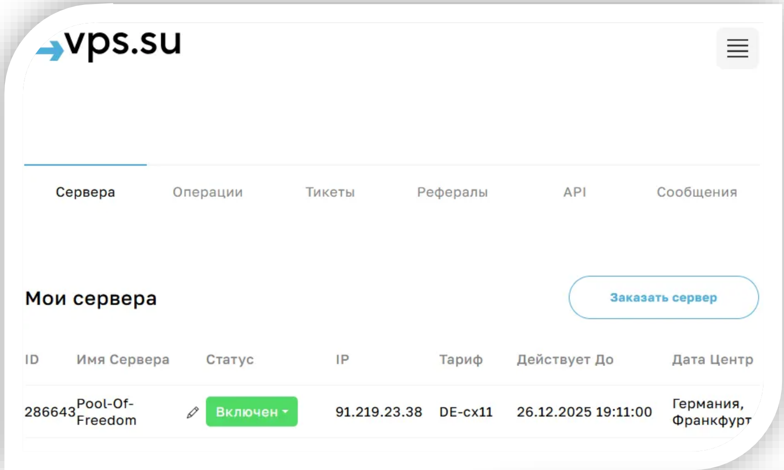
Nikita Niakhai/M25-SNE-01/Unix-based Applications

Secure email server setup in the production

Abstract: This project presents the design, deployment, and validation of a secure email server infrastructure based on Postfix and Dovecot on a pruduction Linux virtual private server with a real domain.

Steps

1. System Preparation and Initial Setup
2. Postfix SMTP Server Installation and Configuration
3. Dovecot IMAP/POP3 Server Setup
4. TLS/SSL Encryption & Email Authentication Configuration
5. Email Client Setup Using Thunderbird
6. SpamAssassin for blocking spam
7. Troubleshooting methods to solve errors
8. Production



4VPS.SU

- Location: Germany;
- 1 CPU, AMD EPYC 7763, 3.3 GHz;
- 1 GB RAM, 10 GB NVMe ;
- 2Gbit/s.

TIMEWEB

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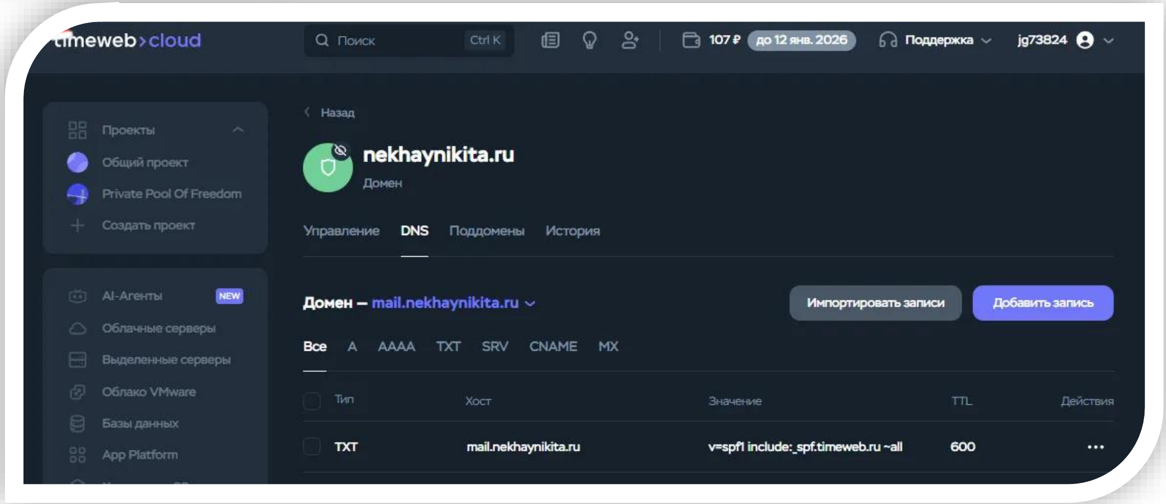

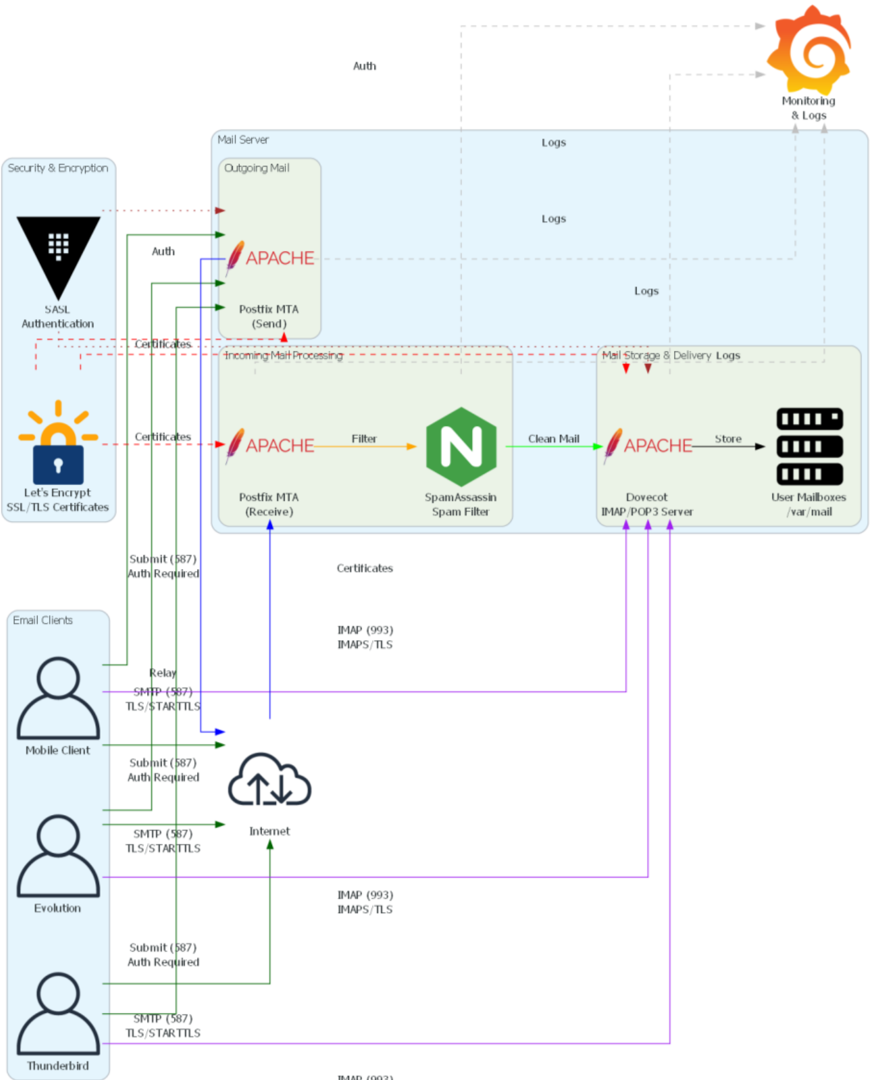

Model

- VPS-based deployment with a dedicated mail subdomain (mail.domain.tld).
- Postfix handles SMTP submission and delivery; Dovecot provides IMAP access.
- SpamAssassin integrated into the mail flow before final delivery.
- DNS, TLS, and authentication services tightly coupled to mail processing.

DNS RECORDS

A
(Mail Host Address Mapping)

Maps the mail server hostname to the public IPv4 address of the VPS.

MX
(Mail Exchange Routing)

- All inbound mail relies on MX resolution.
- Priority determines fallback behavior.

DKIM
(DomainKeys Identified Mail)

- Proves message integrity.
- Confirms domain ownership.
- Mandatory for DMARC alignment.

DMARC

- Enforces SPF/DKIM policy (how receiving servers should handle authentication failures).
- Enables monitoring via aggregate reports.
- Required for modern email trust.

SPF

- Prevents sender address spoofing (which servers are authorized to send email for the domain.).
- Required by Gmail and most modern MTAs.
- First authentication check performed.

PTR
(Reverse DNS)

- Major providers (Gmail, Outlook) heavily penalize missing PTR.
- Required for SMTP reputation and spam scoring.
- Must exactly match the SMTP banner hostname.

Configuration postfix

```
# ------------------------------
# Identity and basic parameters
# ------------------------------
myhostname = mail.nekhaynikita.ru
mydomain = nekhaynikita.ru
myorigin = $mydomain
mydestination = localhost
inet_interfaces = all
inet_protocols = ipv4

# ------------------------------
# Mail storage configuration
# ------------------------------
home_mailbox = Maildir/
mailbox_size_limit = 0
recipient_delimiter = +

# ------------------------------
# Virtual mailbox configuration
# ------------------------------
virtual_mailbox_domains = nekhaynikita.ru
virtual_mailbox_base = /var/mail/vhosts
virtual_mailbox_maps = hash:/etc/postfix/vmailbox
virtual_minimum_uid = 100
virtual_uid_maps = static:5000
virtual_gid_maps = static:5000

# ------------------------------
# SMTP relay and restrictions
# ------------------------------
smtpd_relay_restrictions =
    permit_mynetworks,
    permit_sasl_authenticated,
    defer_unauth_destination
```

```
# ------------------------------
# SASL authentication
# ------------------------------
smtpd_sasl_auth_enable = yes
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth
smtpd_sasl_security_options = noanonymous
smtpd_sasl_local_domain = $myhostname
broken_sasl_auth_clients = yes

# ------------------------------
# TLS configuration (Let's Encrypt)
# ------------------------------
smtpd_use_tls = yes
smtpd_tls_cert_file = /etc/letsencrypt/live/mail.nekhaynikita.ru/fullchain.pem
smtpd_tls_key_file = /etc/letsencrypt/live/mail.nekhaynikita.ru/privkey.pem
smtpd_tls_security_level = may
smtpd_tls_auth_only = yes
smtpd_tls_loglevel = 1
smtpd_tls_received_header = yes
smtpd_tls_session_cache_timeout = 3600s

smtp_tls_security_level = may
smtp_tls_loglevel = 1

# ------------------------------
# DKIM (OpenDKIM integration)
# ------------------------------
milter_default_action = accept
milter_protocol = 6
smtpd_milters = inet:localhost:8891
non_smtpd_milters = inet:localhost:8891

# ------------------------------
# SpamAssassin integration
# ------------------------------
content_filter = spamassassin
```

Configuration TLS/SSL

`/etc/dovecot/conf.d/10-ssl.conf` ;

`main.cf`

```
# SSL/TLS support: yes, no, required. <doc/wiki/SSL.txt>
ssl = required

# PEM encoded X.509 SSL/TLS certificate and private key. They're opened before
# dropping root privileges, so keep the key file unreadable by anyone but
# root. Included doc/mkcert.sh can be used to easily generate self-signed
# certificate, just make sure to update the domains in dovecot-openssl.cnf
ssl_cert = </etc/letsencrypt/live/mail.nekhaynikita.ru/fullchain.pem
ssl_key = </etc/letsencrypt/live/mail.nekhaynikita.ru/privkey.pem
```

```
for p in 25 587 465 143 993 110 995 80 443; do sudo ufw allow ${p}/tcp;
done && sudo ufw enable
```

```
sudo apt install -y certbot
sudo certbot certonly --standalone -d mail.example.com
```

```
root@139679: # ls -l /etc/letsencrypt/live/mail.nekhaynikita.ru/
total 4
-rw-r--r-- 1 root root 692 Dec 16 12:59 README
lrwxrwxrwx 1 root root  44 Dec 16 12:59 cert.pem -> ../../archive/mail.nekhaynikita.ru/cert1.pem
lrwxrwxrwx 1 root root  45 Dec 16 12:59 chain.pem -> ../../archive/mail.nekhaynikita.ru/chain1.pem
lrwxrwxrwx 1 root root  49 Dec 16 12:59 fullchain.pem -> ../../archive/mail.nekhaynikita.ru/fullchain1.pem
lrwxrwxrwx 1 root root  47 Dec 16 12:59 privkey.pem -> ../../archive/mail.nekhaynikita.ru/privkey1.pem
```

Configuration User Authentication SASL

```
CLIENT SIDE
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Thunderbird
  ├─ SMTP server: mail.domain.tld
  ├─ Port: 587
  ├─ Security: STARTTLS
  └─ Auth: Normal password (LOGIN / PLAIN)


SERVER SIDE
══════════════════════════════════════════════════════

┌──────────────────────────────────────────────────────┐
│ Postfix (SMTP submission service)          │
│ /etc/postfix/master.cf                     │
│                                            │
│ submission inet ... smtpd                  │
│  - smtpd_sasl_auth_enable=yes              │
│  - smtpd_tls_security_level=encrypt        │
│                                            │
│ /etc/postfix/main.cf                       │
│  - smtpd_sasl_type = dovecot               │
│  - smtpd_sasl_path = private/auth          │
│  - smtpd_sasl_security_options = noanonymous │
└──────────────┬───────────────────────────────────────┘
               │
               │ UNIX socket (secure, local)
               │
               ▼
┌──────────────────────────────────────────────────────┐
│ Dovecot Authentication Service             │
```

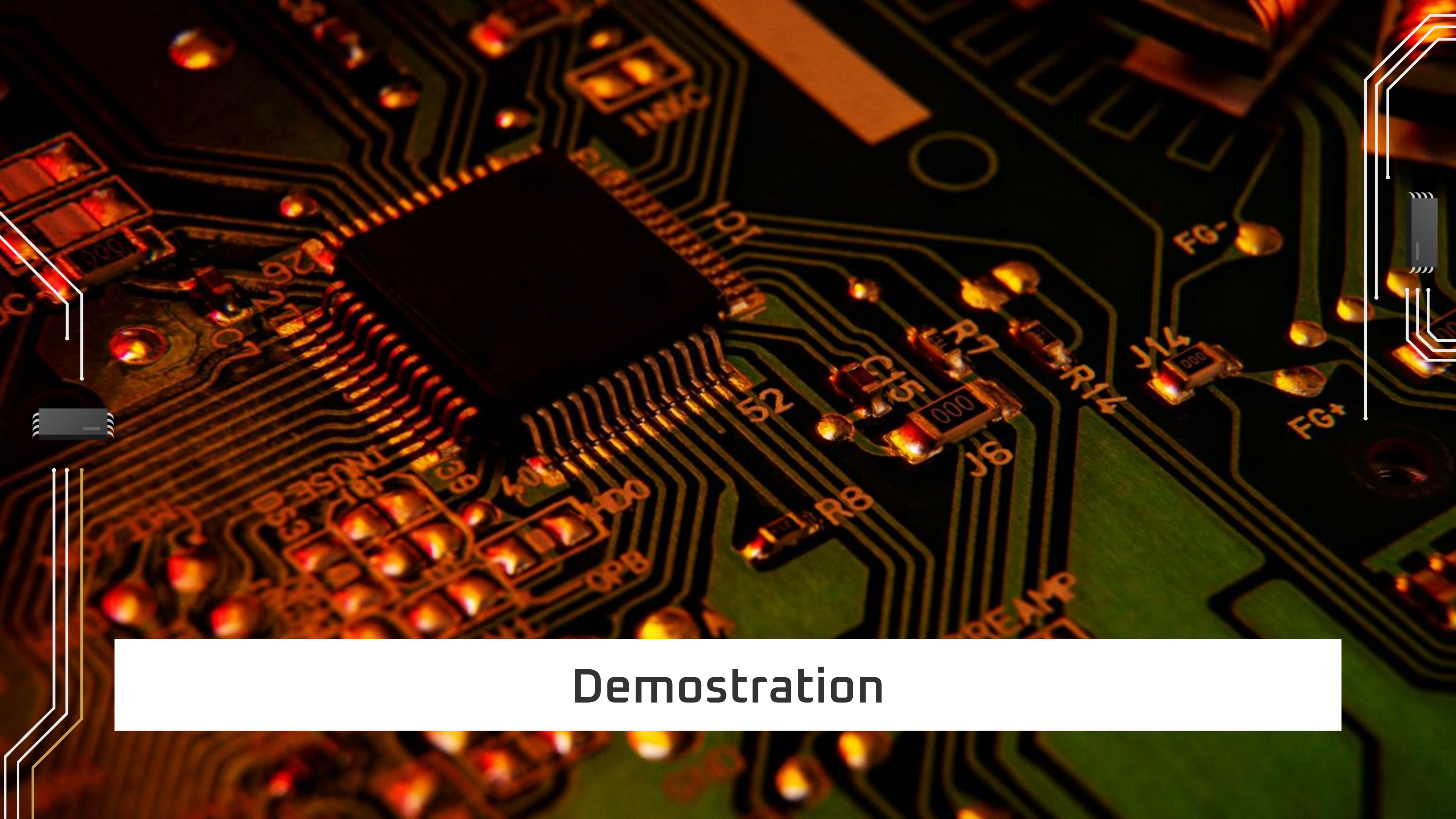
```
          │
          ▼
┌──────────────────────────────────────────────────────┐
│ Dovecot Authentication Service               │
│ /etc/dovecot/conf.d/10-master.conf           │
│                                              │
│ service auth {                              │
│   unix_listener /var/spool/postfix/private/auth │
│ }                                            │
└──────────┬───────────────────────────────────────────┘
          │
          │ AUTH request (username/password)
          ▼
┌──────────────────────────────────────────────────────┐
│ Dovecot auth backend                         │
│ /etc/dovecot/conf.d/10-auth.conf             │
│                                              │
│ !include auth-passwdfile.conf.ext           │
└──────────┬───────────────────────────────────────────┘
          │
          │ lookup
          ▼
┌──────────────────────────────────────────────────────┐
│ Virtual users database                       │
│ /etc/dovecot/users                           │
│                                              │
│ user@domain.tld:password                     │
└──────────────────────────────────────────────────────┘
```

Configuration spamassassin

```
Incoming SMTP mail
        |
        ▼
┌──────────────────────┐
| Postfix (smtpd)      |
| port 25 / 587        |
| main.cf              |
└──────────────────────┘
          | content_filter=spamassassin
          ▼
┌──────────────────────┐
| Postfix pipe         |
| master.cf            |
| spamassassin         |
└──────────────────────┘
          | spamc (client)
          ▼
┌──────────────────────┐
| spamd daemon         |
| SpamAssassin engine  |
| rule + Bayes         |
└──────────────────────┘
          | scored message
          ▼
┌──────────────────────┐
| Postfix reinject     |
| sendmail             |
└──────────────────────┘
          |
          ▼
┌──────────────────────┐
| Dovecot (IMAP)       |
| Maildir delivery     |
└──────────────────────┘
```

```
Message arrives
   ↓
Rule checks (headers, body, URLs)
   ↓
Bayesian probability calculation
   ↓
Total score computed
   ↓
Score ≥ required_score ?
   ├─ YES → mark as SPAM
   └─ NO  → deliver normally
```

Postfix routes incoming mail through SpamAssassin using a pipe transport,where spamc submits messages to the spamd daemon for scoring before themail is reinjected and delivered to the user mailbox.



Demostration

LIMITATIONS

LIMITATIONS

- **Email deliverability challenges**: VPS IPs often face blacklisting or strict scrutiny from providers like Gmail, even with SPF/DKIM/DMARC, due to shared hosting reputation and lack of warm-up. Minor misconfigurations (e.g., DNS mismatches or forwarding) can cause intermittent failures, as seen in the report's troubleshooting.
- **Outdated spam filtering effectiveness**: SpamAssassin relies on rule-based and basic Bayesian analysis, which is less effective against modern adaptive spam (AI-generated or evasive tactics) compared to dynamic systems like those from Gmail. It requires ongoing manual updates and can produce false positives/negatives on low-resource hardware.
- **Security and maintenance burden**: Self-managed open-source components demand constant updates, monitoring, and troubleshooting (e.g., PID errors, loops, or certificate renewals), with risks from misconfigurations or unpatched vulnerabilities. The single-server setup lacks redundancy, making it vulnerable to downtime from failures or attacks.

Refferences

- **Spamassassin installation** https://arenda-server.cloud/blog/kak-nastroit-pochtovyj-server-postfix-s-dovecot/
- **Postfix Official Documentation.** https://www.postfix.org/documentation.html
- **Dovecot IMAP Server Documentation.** https://doc.dovecot.org/
- **RFC 7208 – Sender Policy Framework (SPF).**
- https://datatracker.ietf.org/doc/html/rfc7208
- **RFC 7489 – Domain-based Message Authentication, Reporting, and Conformance (DMARC).** https://datatracker.ietf.org/doc/html/rfc7489
- **Mozilla Thunderbird Documentation.** https://support.mozilla.org/en-US/products/thunderbird

Thanks for the attention!

Any questions?